

# **COURSE MATERIAL**

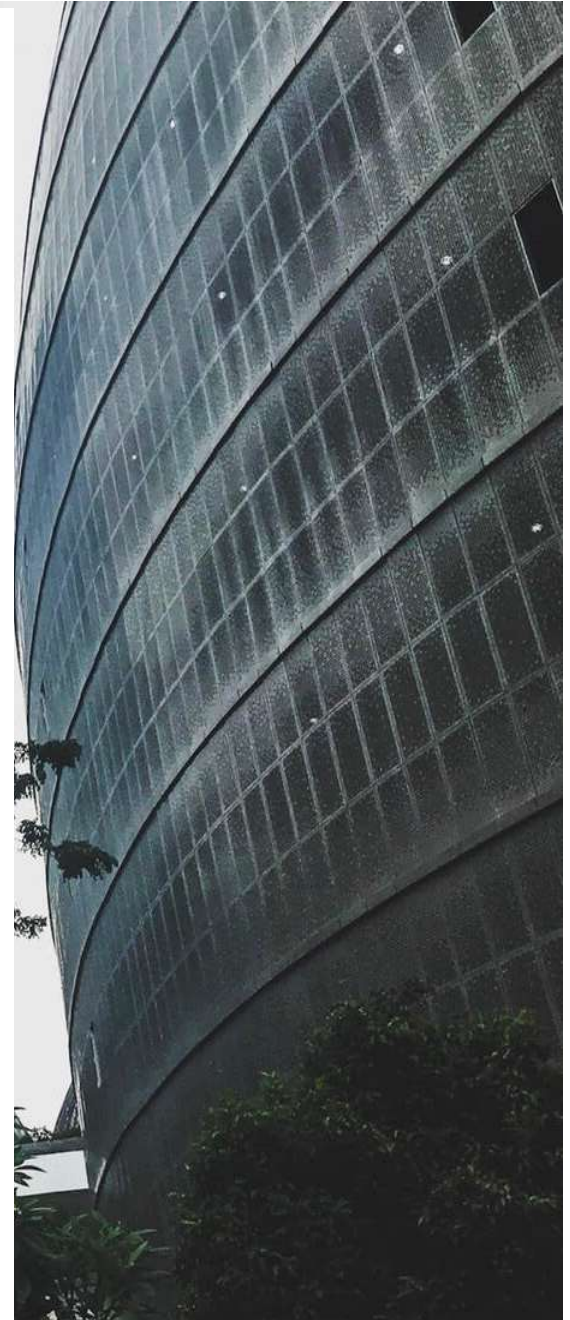
**IF670 – Mobile Cross Platform Programming  
S1 INFORMATICS GRADUATE PROGRAM  
FACULTY OF ENGINEERING AND INFORMATICS**

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**INFORMATICS STUDY PROGRAM  
FACULTY OF ENGINEERING AND INFORMATION  
UNIVERSITAS MULTIMEDIA NUSANTARA**

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# MODUL 10

## (Maps & Geolocation)

### THEME DESCRIPTION

Students are able to implement **geolocation features** in an Expo application.

### WEEKLY LEARNING GOAL (SUB-GOAL LEARNING)

Students are able to utilize **native features** such as **geolocation and maps** in application development..

### SOFTWARE / HARDWARE REQUIREMENTS

1. Laptop / PC
2. IDE (Android Studio/XCode)
3. Code Editor (Visual Studio Code/Sublime/etc)

### PRACTICUM STEPS

1. Install the required dependencies "npx expo install expo-location react-native-maps"
2. Add a "Get Geo Location" button that will request location permission to retrieve the user's coordinates.
3. Display a map to obtain the latitude and longitude.

```
1  import * as Location from "expo-location";
2  import React, { useState } from "react";
3  import { Button, Dimensions, StyleSheet, Text, View } from "react-native";
4  import MapView, { Marker, Region, UriTile } from "react-native-maps";
5
6  type Coordinates = {
7    latitude: number;
8    longitude: number;
9  };
10
11 // Get the device's screen height for responsive map sizing
12 const { height } = Dimensions.get("window");
13
14 export default function App() {
15   const [location, setLocation] = useState<Coordinates | null>(null);
16
17   //async function to get the user's current location
18   const getLocation = async (): Promise<void> => {
19     const { status } = await Location.requestForegroundPermissionsAsync();
20
21     if (status !== "granted") {
22       alert("Permission denied! Please allow location access.");
23       return;
24     }
25
26     const loc = await Location.getCurrentPositionAsync({});
27     setLocation({
28       latitude: loc.coords.latitude,
29       longitude: loc.coords.longitude,
30     });
31   };
32 }
```

```

33 // Define the region for the map based on the user's location
34 const region: Region | undefined = location
35 ? {
36     latitude: location.latitude,
37     longitude: location.longitude,
38     latitudeDelta: 0.01,
39     longitudeDelta: 0.01,
40 }
41 : undefined;
42
43 return (
44     <View style={styles.container}>
45         {!location ? (
46             <Button title="Get Geo Location" onPress={getLocation} />
47         ) : (
48             <>
49                 <MapView style={styles.map} initialRegion={region}>
50                     { /* OpenStreetMap Tile */ }
51                     <UrlTile urlTemplate="https://tile.openstreetmap.org/{z}/{x}/{y}.png" />
52
53                     { /* Marker */ }
54                     <Marker coordinate={location} title="My Location" />
55                 </MapView>
56
57                 <View style={styles.info}>
58                     <Text>Latitude: {location.latitude}</Text>
59                     <Text>Longitude: {location.longitude}</Text>
60
61                     <Button title="Refresh Location" onPress={getLocation} />
62                 </View>
63             </>
64         )}
65     </View>
66 );
67 }

```

```

69 const styles = StyleSheet.create({
70     container: {
71         flex: 1,
72     },
73     center: {
74         flex: 1,
75         justifyContent: "center",
76         alignItems: "center",
77     },
78     map: {
79         height: height * 0.5,
80         width: "100%",
81     },
82     info: {
83         flex: 1,
84         padding: 16,
85         backgroundColor: "■ #fff",
86     },
87 });

```

## ASSIGNMENT

After successfully obtaining the current location, enhance your application by adding the following features:

- Make the marker on the map movable
- The marker position can be changed by:
  - o tapping on the map, or
  - o dragging the marker

Each time the marker position changes, the latitude and longitude values must be updated immediately and displayed on the screen!

## REFERENSI